

Reserve Pathways

Problem ID: reservepathways-27-24174788

A company has been contracted to create a connected set of pathways for a reserve between major points of interest. They are quite greedy, so they want to add the cheapest sets of roads possible to ensure that all points of interest remain connected. The company has specifically entrusted you with the task of saving as much money as possible; they have provided you with a plan proposed by a *previous* landscape architect, depicting a connected network. Additionally, the company may construct any of the remaining paths from the plan, provided that the total cost does not exceed T . They would like you to report the total cost required to establish the cheapest-possible connected network, along with the maximum number of roads which can be built from the old plan.

Input

- The first line contains 3 integers $1 \leq V \leq 10^5$, the number of points of interest, $V - 1 \leq E \leq 2 \times 10^5$, the total number of paths, and $0 \leq T \leq \sum_{i=1}^E w_i$, where the upper bound denotes the sum of the weights of all edges, the allowed cost to construct additional roads from the given network.
- Each other line contains 3 integers u, v, w , where $0 \leq u \leq V - 1$, $0 \leq v \leq V - 1$, and $u \neq v$, which are points of interest, and $w \geq 0$, which is the price required to establish a path from u to v .

Output

Output two integers: the total cost of the network, and the maximum number of paths which can be added to that network.

Sample Input 1

```
5 6 12
0 1 5
0 2 3
1 2 2
2 3 7
3 4 5
1 4 3
```

Sample Output 1

```
13 2
```

Sample Input 2

```
4 4 0
0 1 1
0 1 2
1 2 1
2 3 1
```

Sample Output 2

```
3 0
```